The forecast saw it coming 6 days before Malibu went dark.

PoweruP · fleet command for home batteries, before and during a shutoff

Dual-Use Hackathon · Energy track with Base Power

BRYSON FIRE · REPLAY

Oct 9, 2:14 AM
NWS first mentions possible Santa Ana winds
NWS REAL

> **WATCH** INFERRED
> Flag high-risk circuits; check which batteries report in.

Oct 12
SCE tells the City it may cut Cuthbert, Galahad, Maguire, Merlin from Oct 15
SCE / City of Malibu REAL

> **ASK** INFERRED
> Text residents; get Medical Baseline consent to hold batteries at 100%.

Oct 13
Fire Weather Watch issued
NWS REAL

Oct 14, 5:00 PM
Serra and Tuna added (6 circuits); 5 PM Red Flag Warning; heat advisory up to 95°F
SCE / City of Malibu / NWS REAL

> **FILL** INFERRED
> Staggered charging, Medical Baseline homes first; export paused.

Oct 15, 4:41 PM
Bryson Fire starts 4:41 PM; power out in central/west Malibu by 5 PM
Cal Fire / City of Malibu / SCE REAL

> **HOLD** INFERRED
> Homes island; each gets an hours budget.

Oct 16, 3:00 AM
3 AM: four batteries stop reporting
PoweruP scenario ASSUMED

> **HOLD** INFERRED
> Silent batteries keep their local rule; classified not talking / tripped / broken.

Oct 16, 7:00 AM
69,746 SCE customers out (LA + Ventura counties)
SCE REAL

After the winds
After the winds: restoration takes 8+ hours
SCE public reports REAL

> **HAND OFF** INFERRED
> Propose a staggered recharge for the SCE shutoff officer to approve, avoiding a cold-start surge.



Six circuits de-energized: Cuthbert, Galahad, Merlin, Maguire, Serra, Tuna · ≈6,374 customers on the five with a public count REAL · Outline: 2020 Census ZCTA REAL

Hours of backup per home MIXED

48 h

with a per-home hours budget vs 28 h for a full Base Core (39.2 kWh) with no plan, at Malibu's real average load (1.28 kW, SCE 2024-06)

Homes kept on through 48 h MIXED

6 of 6

2 of 2 Medical Baseline homes protected · 0 of 6 without a plan

Peak headroom freed ASSUMED

10 kW

staggered charging vs everyone at once (18 kW) · restoration surge 26 → 16 kW · ≈0.83 MW at a 500-home pilot

TWO CUSTOMERS, ONE FLEET

1 · Ordinary Tuesday

SCE peak relief at 6 PM from energy above each home's reserve

$9.06 per 2 h event, 6 homes
ASSUMED prices

2 · Shutoff day

Utility DR dispatcher and shutoff officer: staggered fill, per-home budgets, silent-battery triage, staggered recharge for **human approval**.

FUNDING

DOE SBIR/STTR primary

Reauthorized April 2026 through 2031. Fit: grid resilience and distributed energy resources.

Phase I	500-home SCE pilot on PSPS-prone circuits
Phase II	5,000 homes across multiple circuits, utility DR integration
Phase III	Utility / aggregator contracts: peak relief on ordinary days

Alt: DoD SBIR (AFWERX or Army): base housing kept powered.

ONE ENGINE, ANY REGION

Los Angeles · wildfire (PSPS) REAL	48 h backup · 6/6 kept on
Australia · bushfire SYNTHETIC	48 h backup · 6/6 kept on
Japan · typhoon SYNTHETIC	72 h backup · 6/6 kept on
Community battery · clinics & cell towers SYNTHETIC	72 h backup · 6/6 kept on

Same engine code; only config/regions/*.json changes. Non-LA data is SYNTHETIC.

enqiwang.github.io/powerup/?demo=1
TEAM NAME 1 · TEAM NAME 2 · TEAM NAME 3

REAL public data (NWS, SCE/ORNL, Census, City of Malibu, Cal Fire) · INFERRED what PoweruP would have done · ASSUMED battery state, consent, headroom, prices · SYNTHETIC non-LA regions. Base Core 39.2 kWh; power rating assumed. Simulation only: no real Base customers, no messages sent.